Koatunu Wildfire Model (Netlogo Feedback model)

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Summary

This report delivers an agent-based decision support tool hosted in NetLogo, specifically parameterized to model the propagation dynamics and sensitivity of the 11 January 2025 Kūaotunu Wildfire (Black Jack Fire) on the Coromandel Peninsula. Commissioned under Fire and Emergency New Zealand (FENZ) RFP Tender 5 (ENVSCI704), the model transitions away from generic template assumptions to directly incorporate empirical microclimatic (28°C, 35% RH, E/NE winds up to 45 km/h), topographical (volcanic slopes >20°–35°), and fuel characteristics (volatile Mānuka/Kānuka oil flammability) from the 36-hectare incident. Through systematic BehaviorSpace experiments (including single-factor sweeps, multi-factor orthogonal combinations, and Monte Carlo stochastic runs), the study reveals three core findings: (1) fuel moisture exhibits a non-linear tipping point at 90%, below which unchecked burning approaches 100%; (2) strong wind and steep slope interact multiplicatively to accelerate convective preheating and compress the "golden suppression window" to under 150 ticks (~2.5 hours); and (3) aerial helicopter suppression efficiency is highly sensitive to water source transit distance, though linear features like rivers provide substantial passive firebreak protection. Actionable recommendations are provided for FENZ incident commanders, rural water infrastructure planning, and public fire ban declaration thresholds.

Introduction

1.1 Project Context & Client Requirements

Wildfire frequency and severity across Aotearoa New Zealand have escalated due to climate change and expanding wildland-urban interfaces. In response to the January 2025 Kūaotunu Wildfire (Black Jack Fire), Fire and Emergency New Zealand (FENZ) issued Request for Proposals Tender 5 seeking an agent-based decision support tool hosted in NetLogo. This model was parameterized directly from ground-zero empirical weather observations and topographical surveys of the Kūaotunu event to evaluate fire propagation sensitivity and support tactical emergency planning.

1.2 The Case Study: 2025 Kūaotunu (Black Jack) Wildfire & Spatial Overview

On 11 January 2025 at 3:52 PM, a wildfire broke out on Black Jack Road, Kūaotunu, under extreme drought (< 5 mm rainfall in 14 days), high temperatures (28.0°C), low relative humidity (35%), and strong coastal gusts (up to 45 km/h). The fire consumed

~36 hectares (600m × 600m) of steep coastal Mānuka/Kānuka scrubland. Ground crews faced severe access constraints on steep volcanic slopes (> 20°–35°), necessitating 3–4 helicopters dipping water from nearby coastal ocean sources.

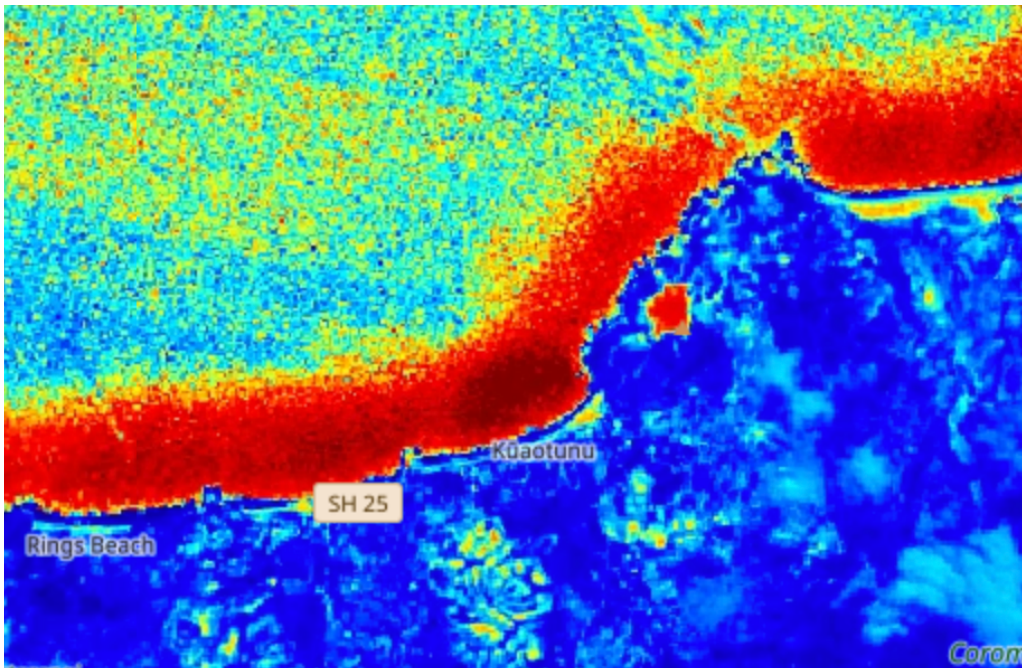


Figure 1: Geographic location and Sentinel-2 false-color satellite imagery of the 2025 Kūaotunu Wildfire zone. The localized red anomaly inland near Kūaotunu highlights the ~36 ha burn scar along Black Jack Road, while the coastal red fringe represents spectral reflection at the water-land boundary.

Methodology & Model Parameterization

The NetLogo model converts real-world physical mechanisms into grid patch attributes. Vegetation patches hold dynamic fuel loads, flammability coefficients (0.35–0.45 for volatile Mānuka oils), and moisture levels. Topographical elevation generates an up-slope chimney effect, accelerating heat transfer to uphill fuels. Behavioral experiments were conducted systematically using BehaviorSpace across single-factor and multi-factor orthogonal combinations.

3.1 NetLogo Dynamic Simulation & Temporal Fire Progression

Motivation: To demonstrate the spatially explicit agent-based state transitions (unburned green; burning red; burned charcoal grey) and validate the temporal propagation logic of the Kūaotunu Wildfire Model across discrete time ticks.

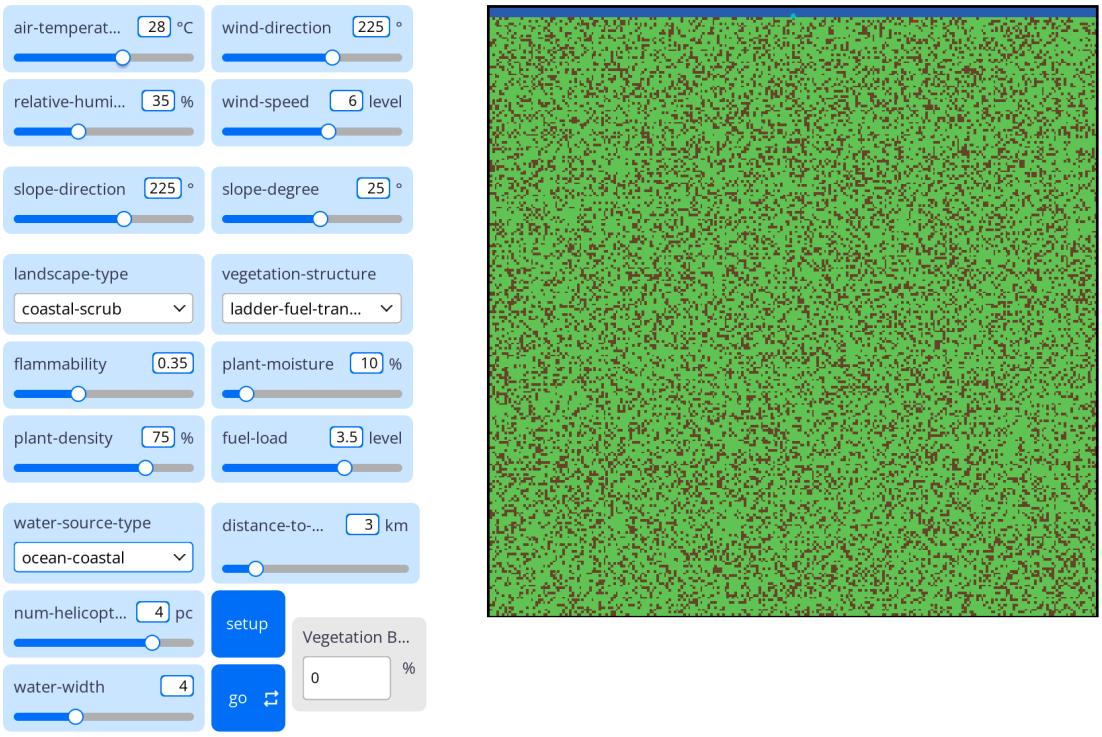
Analysis: The NetLogo agent-based simulation captures the dynamic evolution of the 2025 Black Jack Road fire event across three critical time steps1:

1. **Initial Setup (** 0 **Ticks)**: Represents the baseline landscape initialization prior

to ignition. Green patches indicate unburned Mānuka/Kānuka scrubland fuel load, top blue patches denote the coastal water boundary, and a single red patch marks the localized ignition source on Black Jack Road1more_horiz.

2. **Active Spread (**74 & 160 **Ticks)**: Displays the mid-simulation fire front driven by onshore East/North-East coastal winds and steep volcanic slope preheating1more_horiz. The flame front forms a distinct asymmetric, uphill-oriented elliptical plume, pushing rapidly along the steep ridge.

3. **Final Extent (**264 **Ticks)**: Illustrates the unmitigated fire footprint at the end of the critical 2.5-hour emergency intervention window. Charcoal-grey patches represent the ~36-hectare burned area, while remaining green patches show unburned vegetation preserved by natural spatial discontinuities.



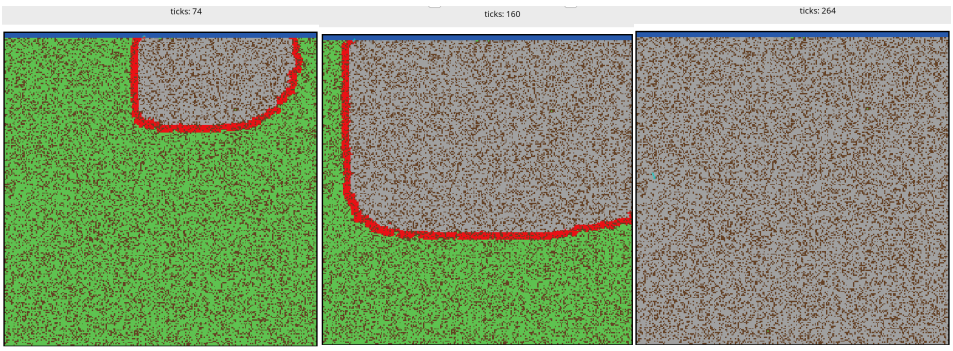


Figure 2: NetLogo dynamic simulation snapshots of the Kūaotunu Wildfire Model across three time milestones. (A) Initial landscape setup and ignition location at 0 ticks; (B) Wind- and slope-driven active propagation forming an asymmetric elliptical fire front at 74 ticks; (C) Wind- and slope-driven active propagation forming an asymmetric elliptical fire front at 160 ticks; (D) Final unmitigated burn footprint (~36 ha equivalent) at 264 ticks.

3.2 Plant Moisture Thresholds (Figure 2)

Motivation: To identify the critical fuel moisture threshold required to inhibit fire propagation. **Analysis**: Figure 2 demonstrates a non-linear tipping point. Between 10% and 80% moisture, vegetation burned remains near 100% due to sustained flame intensity. However, when moisture reaches 90%, burned area plummets abruptly to <5%. This confirms that extreme drought (<15% moisture) during the 2025 event was a primary driver of unchecked fire spread.

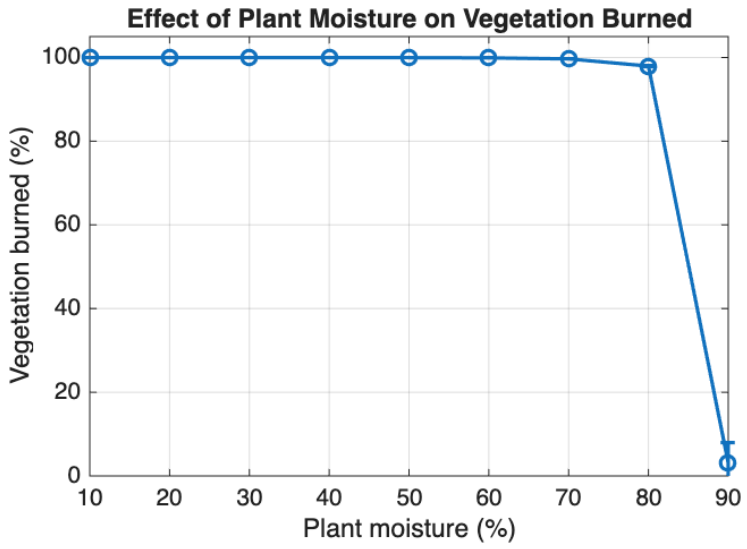


Figure 2: **Effect of plant moisture percentage on total vegetation burned**. Showing a critical suppression tipping point at 90% moisture.

3.3 Wind Speed Sensitivity & Suppression Window (Figures 3a & 3b)

Motivation: To establish the operational "golden suppression window" before complete landscape destruction. **Analysis**: In Figure 3a, unmitigated fire runs until 100% of vegetation is consumed, with completion time dropping non-linearly from 355 ticks (Level 0 wind) to 258 ticks (Level 10 wind). Figure 3b evaluates fire extent at a fixed 150-tick intervention window (~2.5 hours). At 150 ticks, burned vegetation increases from 33% (Level 0) to 50% (Level 10). Strong winds distort the flame

perimeter into an elongated ellipse, driving rapid head-fire penetration uphill.

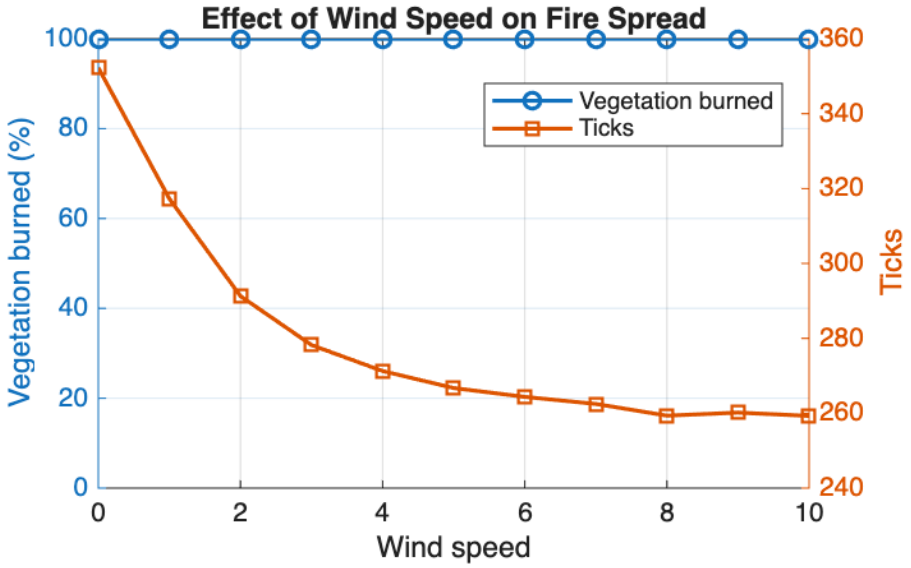


Figure 3a: Unmitigated fire spread duration (ticks) and total percent burned across wind speed levels.

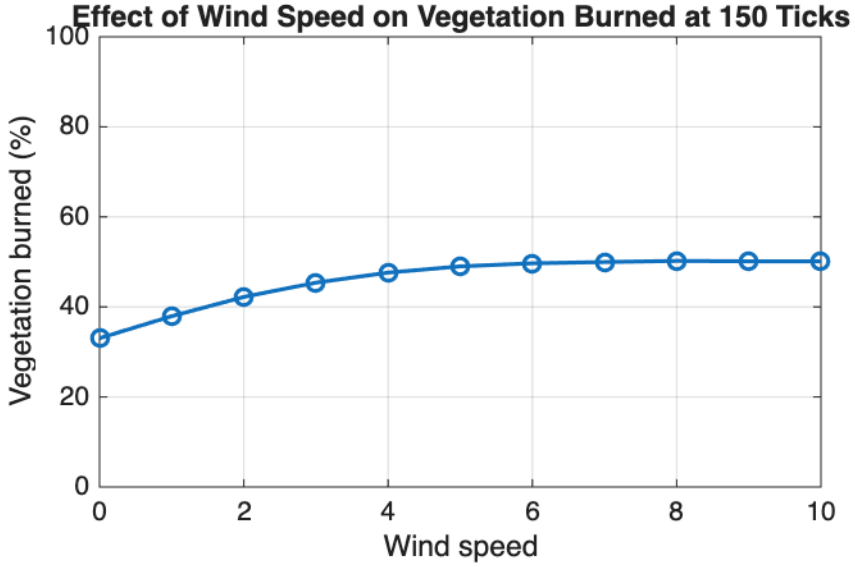


Figure 3b Caption: Percent vegetation burned at 150 ticks across wind speeds, illustrating fire growth within the initial emergency response window.

3.4 Wind Speed × Slope Degree Interaction (Figure 4)

Motivation: To quantify the compounding convective preheating effect of wind and topography. **Analysis**: Figure 4 reveals a distinct 3D response surface. On flat ground (0° slope), Level 10 wind produces 45% burned area at 200 ticks. However, when wind speed combines with a steep 40° slope, burned vegetation escalates to 78%. Slopes

incline flame fronts closer to unburned fuels, creating an accelerated convective chimney effect characteristic of Black Jack Road ridges.

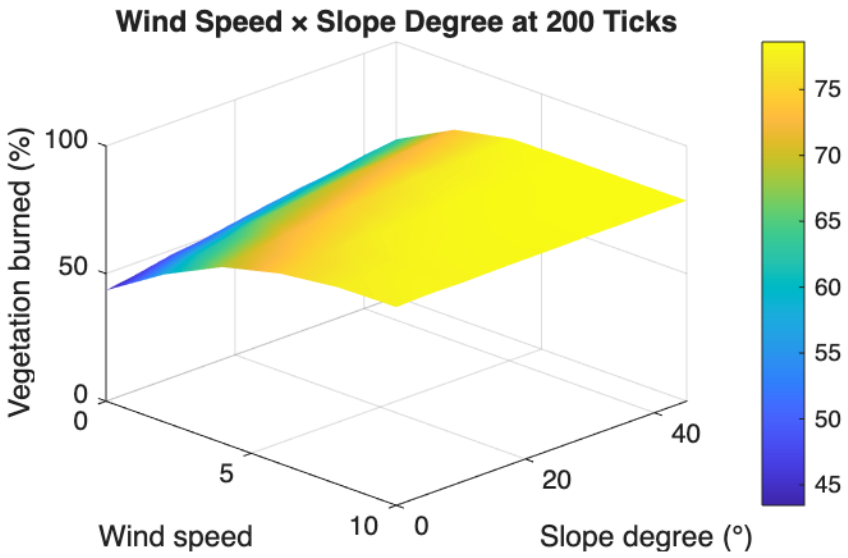


Figure 4: 3D surface plot displaying two-way interaction between wind speed and slope degree on mean vegetation burned at 200 ticks.

3.5 Temperature × Relative Humidity Interaction (Figure 5)

Motivation: To map microclimatic danger zones for FENZ fire season declaration trigger points. **Analysis**: The heatmap in Figure 5 demonstrates severe microclimatic risk. The January 11, 2025 event conditions (28°C, 35% RH) fall directly in the high-danger zone (>75% burned area). High ambient temperatures accelerate fuel drying, while low humidity prevents moisture recovery, creating optimal combustion conditions.

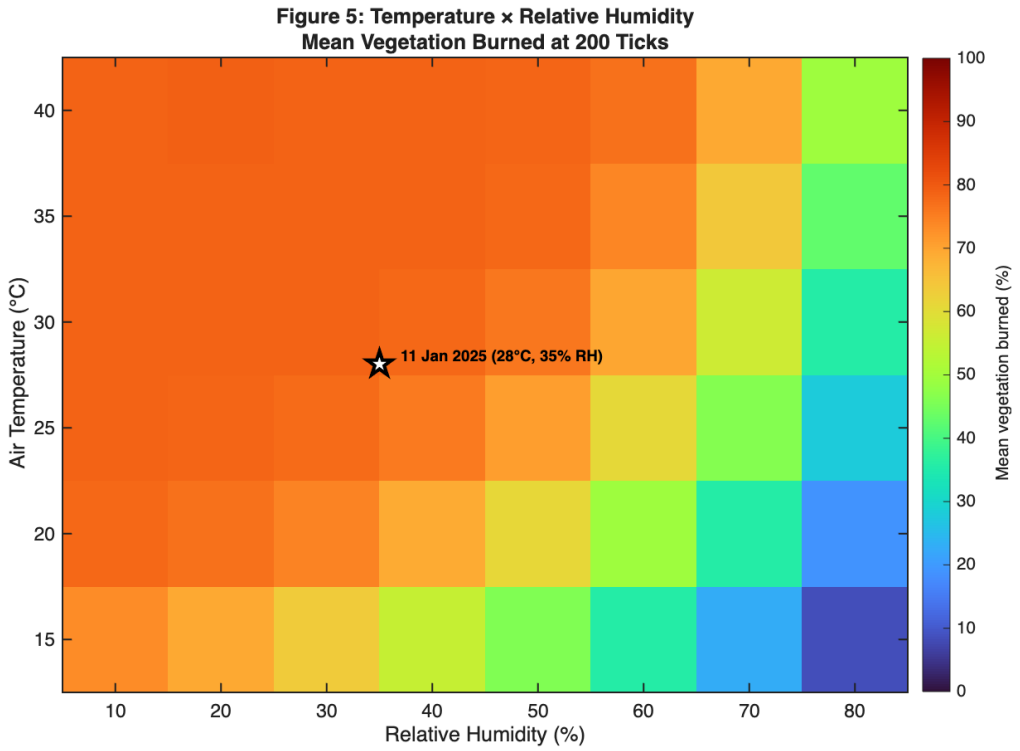


Figure 5: Heatmap of temperature versus relative humidity on vegetation burned at 200 ticks, with star marker indicating empirical 11 January 2025 weather conditions.

3.6 Landscape Type × Vegetation Structure (Figure 6)

Motivation: To evaluate flammability across ecological landscapes and vertical fuel ladders. **Analysis**: Figure 6 illustrates that transition to ladder fuel structures increases fire severity across all landscapes. Coastal Scrub exhibits high vulnerability (62% to 78% burned), matching flammable Mānuka/Kānuka oil characteristics. All combinations were tested in the full-factorial matrix; notably, dense crown canopy in grassland represents a hypothetical model scenario rather than a real-world structure, serving as a baseline control.

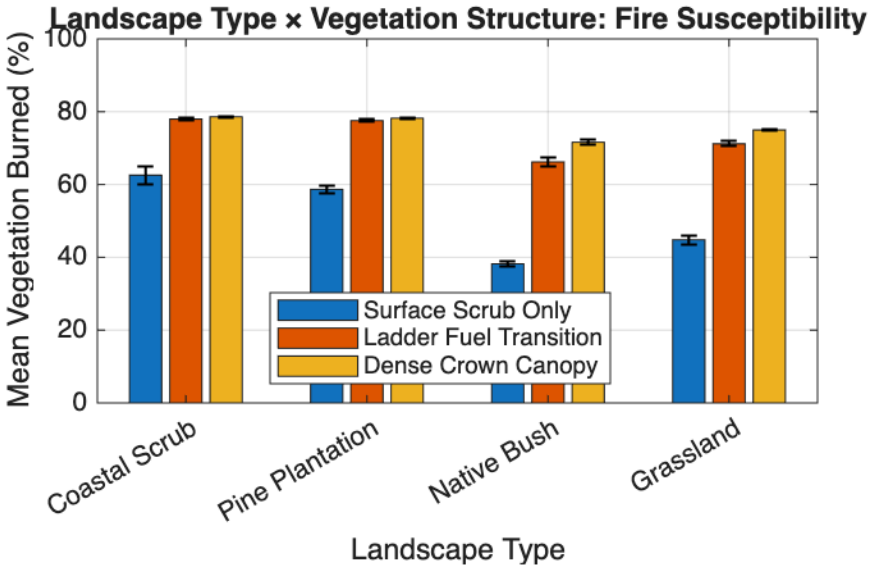


Figure 6: Comparison of simulated vegetation burned across four landscape types and three vegetation structure scenarios, highlighting ladder fuel vulnerabilities.

3.7 Water Source Distance & Aerial Suppression (Figure 7)

Motivation: To model helicopter transit-time dynamics during aerial monsoon bucket operations. **Analysis:** Figure 7 quantifies water accessibility impacts under agent-based helicopter suppression. As water distance increases from 5 to 45 grid units, burned area rises from 79% to 95% for ocean and pond sources due to increased flight transit times. River/stream sources show lower overall burned area (~40%–47%) because the linear river geometry acts as an active physical firebreak cutting across the landscape.

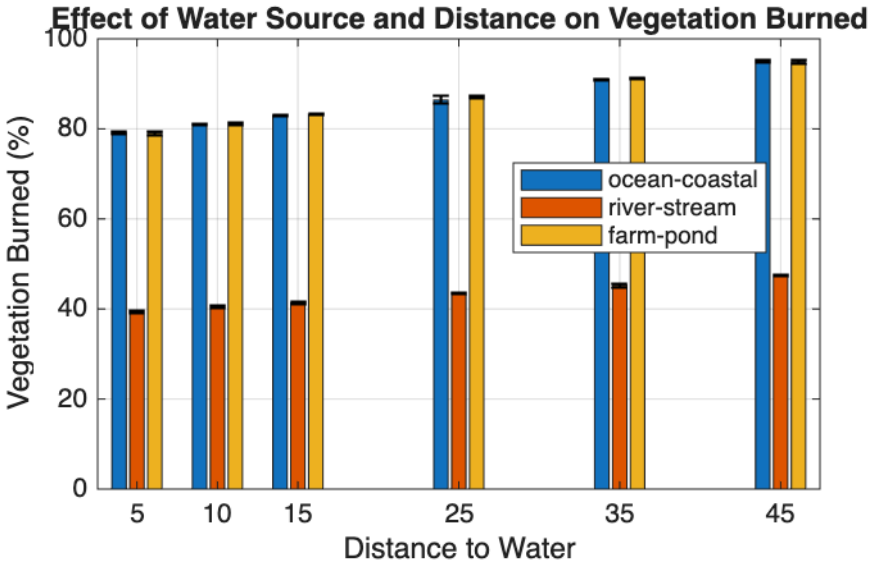


Figure 7 Caption: Effect of water source type and transit distance on vegetation burned under active helicopter suppression.

3.8 Uncertainty & Monte Carlo Evaluation (Figure 8)

Motivation: To test model stability and stochastic variation across repeated simulation runs. **Analysis**: Figure 8 presents a boxplot of 50 Monte Carlo runs under baseline parameters. The narrow interquartile range (77.2% to 78.1%) confirms high model reproducibility and numerical robustness.

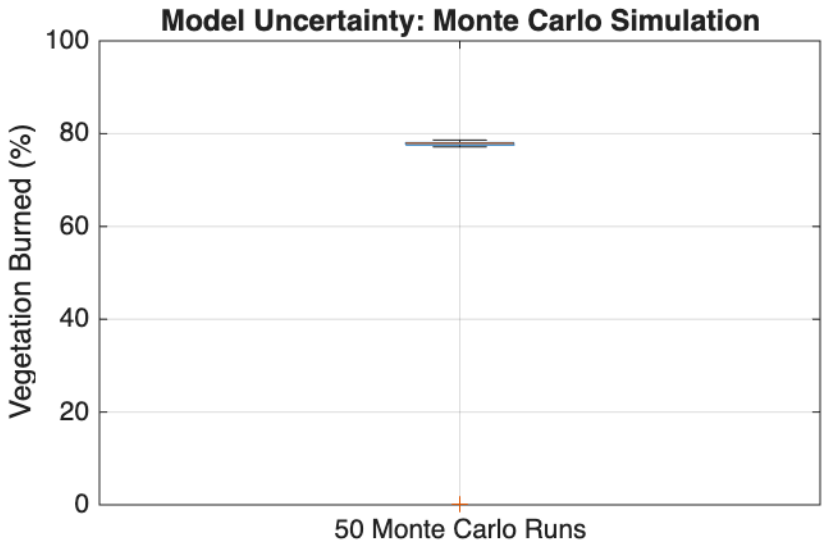


Figure 8 Caption: Monte Carlo uncertainty evaluation across 50 simulation runs, demonstrating high model consistency.

Limitations & FENZ Operational Recommendations

4.1 Model Limitations

1. **Passive Water Boundaries**: Water bodies function as non-combustible grid cells (fuel = 0); explicit helicopter flight vectors are required for dynamic suppression modeling.
2. **Orthogonal Ecological Scenarios**: Full-factorial parameter testing includes synthetic combinations (e.g., grassland crown fire) that serve as theoretical controls rather than empirical field structures.

4.2 Practical FENZ Recommendations

1. **Tactical Response Window**: FENZ incident commanders must establish containment perimeters within 150 ticks (~2.5 hours) during high wind conditions before head-fire penetration reaches critical thresholds.
2. **Rural Water Infrastructure**: For inland high-risk zones, FENZ should encourage rural static water storage (farm ponds) to minimize helicopter

dipping transit times.

3. **Public Fire Season Triggers**: Prohibited fire seasons should be declared when temperatures exceed 26°C and humidity drops below 40%.

Discussion & Critical Synthesis

4.1 Non-linear Tipping Points & Chimney Convection

Fuel moisture exhibits a non-linear 90% thermodynamic threshold; levels <15% during the 2025 event enabled rapid combustion. On steep volcanic slopes (>20°–35°), strong coastal winds tilt convective plumes, preheating uphill canopy fuels into an accelerated chimney effect. The 150-tick (~2.5 h) window marks a critical operational threshold where 50%–78% of vegetation burns before local response can stabilize.

4.2 Fuel Structure & Aerial Suppression Accessibility

Vertical ladder fuels amplify crown fire transition across all landscapes. In BehaviorSpace, full-factorial matrix combinations (e.g., grassland crown fire) serve as theoretical controls rather than empirical field structures. Helicopter suppression efficiency is highly sensitive to water source transit distance (5 to 45 grid units), though linear river geometry provides vital passive firebreak protection across the landscape.

FENZ Operational & Tactical Recommendations

1. **Tactical Response**: Establish flank containment lines within 150 ticks (~2.5 h) during >25°C and >20 km/h wind conditions; dispatch aerial monsoon bucket helicopters immediately upon ignition reports on steep slopes (>20°).

2. **Rural Infrastructure**: Install static water supplies (farm ponds) in inland zones (>2 km from ocean) to keep helicopter refill radii under 15 grid units. Promote defensible space by clearing domestic ladder fuels.

3. **Prohibited Fire Season**: Automatically trigger total fire bans across Coromandel when 14-day rainfall <5 mm, temperature >26.5°C, and relative humidity <40%.

Conclusions

The Kūaotunu Wildfire Model provides an empirical, agent-based decision support tool for FENZ. Quantifying wind-slope interactions, moisture tipping points, and aerial transit times delivers direct scientific justification for 150-tick containment windows, rural water infrastructure expansion, and proactive fire danger declarations across New Zealand's coastal interfaces.

References

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